

Vision-Assisted Bipedal Locomotion on Challenging Terrains via Terrain-Related Adversarial Motion Priors

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Abstract—Humanoid locomotion on challenging terrains requires reliable perception and terrain-adaptive gait regulation to maintain stability under complex contact conditions. Although recent perceptive locomotion methods have made substantial progress, many rely on explicit terrain representations, auxiliary geometry-related supervision, or multi-stage training pipelines, which can increase system complexity and training cost. This paper presents a lightweight single-stage reinforcement learning framework for vision-assisted humanoid locomotion using only onboard proprioception and low-cost depth sensing. A hierarchical feature extractor learns compact latent representations of robot dynamics and terrain context, which are integrated into a mixture-of-experts actor for terrain-aware behavior modulation. We further introduce a terrain-related adversarial motion prior constructed from flat-ground and stair locomotion demonstrations to encourage terrain-compatible motion patterns within the unified policy. Extensive simulation and real-world experiments on a physical humanoid robot show that the learned policy achieves robust and energy-efficient locomotion across slopes, stairs, elevated platforms, and wide gaps, while maintaining stable foot– terrain interactions in cluttered outdoor scenes.

Index Terms—Humanoid and bipedal locomotion, reinforcement learning, deep learning for visual perception.

I. INTRODUCTION

LEGGED robots, particularly bipedal humanoids, offer distinct advantages for deployment in human-centric and unstructured environments due to their anthropomorphic morphology. With recent advances in learning-based locomotion, deep reinforcement learning (RL) has shown strong capability in enabling blind policies that rely solely on proprioception for both quadruped [1], [2], [3] and bipedal robots [4], [5]. While these approaches exhibit strong robustness on uneven ground, they inherently lack look-ahead environmental awareness. As a result, they struggle with maintaining stability and adapting gait on continuous stairs or other complex terrains, which significantly limits the practical deployment of bipedal robots in real-world environments.

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To fully enable versatile locomotion in challenging environments, integrating exteroceptive sensing such as cameras or LiDAR is essential. These sensors allow robots to infer geometric properties of the surrounding terrain from elevation maps [6] or depth images [7]. Building on such perceptual cues, recent studies have demonstrated impressive indoor and outdoor parkour capabilities on quadruped robots [7], [8], [9], [10] and bipedal robots [11]. In contrast, the intrinsic instability and high degrees of freedom of bipedal robots pose significant challenges for learning perceptive locomotion and achieving sim-to-real transfer.

The diverse and irregular structure of real-world terrains also makes reward design a persistent challenge for RL-based locomotion [12]. Existing perception-driven approaches often rely on extensive hand-crafted reward terms to account for terrain geometry [7], [13], [14] and to regularize limb behavior [15]. Although animal or human demonstrations could substantially alleviate this burden, existing efforts primarily focus on multi-gait learning [12] or anthropomorphic walking [16], [17] on relatively simple terrains, leaving the role of motion priors in complex-terrain bipedal locomotion insufficiently explored.

In addition, introducing perceptual inputs makes the policy more vulnerable to observation noise and increases both hardware requirements and training overhead [18]. Under a single-stage training pipeline, the agent often struggles to discover high-return policies directly from high-dimensional sensory inputs. Teacher-student (TS) frameworks [7], [13], [19] and hierarchical architectures [15], [20] can mitigate this optimization difficulty, but they also introduce additional design complexity and reduce training efficiency. This naturally raises the following question: can a lightweight single-stage framework achieve robust and natural locomotion while retaining sufficient perceptual intelligence in complex environments?

In this paper, we propose a lightweight single-stage framework for vision-assisted humanoid locomotion on complex terrains, using only onboard proprioception and low-cost depth sensing. The framework employs a hierarchical feature extractor to learn compact latent representations of robot dynamics and terrain context, which are further integrated into a Mixture of Experts (MoE) actor for terrain-aware behavior modulation. To encourage human-like and terrain-adaptive motion patterns, we introduce a terrain-related adversarial motion prior (TR-AMP) constructed from flat-ground and stair locomotion

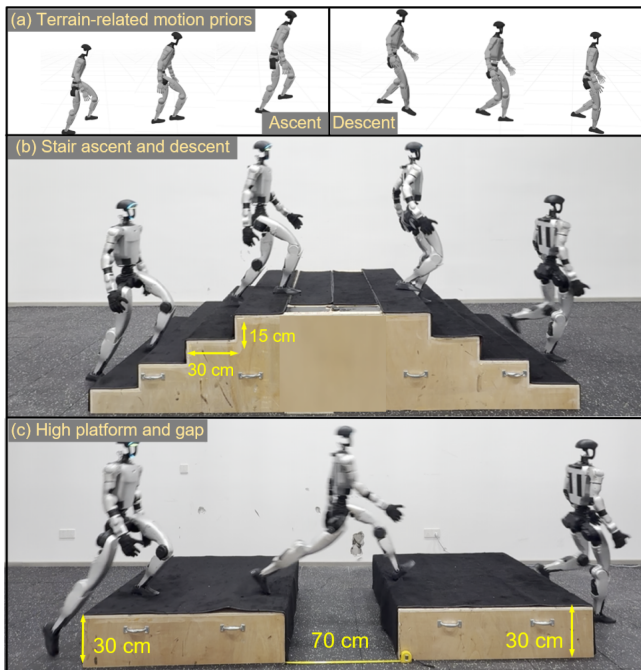


Fig. 1. Our framework enables robust real-world bipedal locomotion on challenging terrains. (a) Terrain-related motion priors used during training. (b) Traversal of a staircase with ascent and descent. (c) Traversal of a 30 cm high platform and a 70 cm gap.

demonstrations. We evaluate the proposed method on the Unitree G1 humanoid robot, where it demonstrates robust real-world locomotion on challenging terrains, including repeated stair traversal, traversal of an elevated platform, and wide-gap crossing, as shown in Fig. 1.

Our main contributions are as follows:

- We present a lightweight single-stage framework that enables effective and robust vision-assisted humanoid locomotion on complex terrains.
- We introduce a terrain-related adversarial motion prior to encourage terrain-adaptive motion patterns within the unified policy, improving body stability, foothold interaction, and locomotion efficiency.
- Physical experiments demonstrate that the learned policy generalizes across diverse terrains, exhibiting robust terrain-aware behaviors and appropriate gait patterns on slopes, stairs, high platforms, and gaps.

II. RELATED WORK

A. Perception-Assisted Humanoid Locomotion

A common way to incorporate exteroception into humanoid locomotion is to use an explicit elevation map [6]. Some recent humanoid locomotion pipelines explicitly build such maps from LiDAR [15], [20], while others reconstruct elevation maps from depth observations with the aid of auxiliary reconstruction objectives [14], [21], [22], [23]. However, the effectiveness of these methods remains strongly tied to the intermediate terrain representation: LiDAR-based map pipelines are sensitive to mapping latency and odometry drift, while reconstruction-based

methods are often difficult to scale to unseen environments and typically require additional supervision or more complex training pipelines.

To reduce reliance on explicit mapping and odometry estimation, depth camera-based approaches have emerged as an alternative. However, learning a unified policy directly from depth remains challenging due to the high-dimensional visual input, limited underfoot observability, and the difficulty of stable policy optimization. As a result, many existing methods still rely on auxiliary supervision [10], [18], teacher-student distillation [11], [19], or multi-stage training [24], which adds training overhead and complexity to the perception pipeline. This motivates our exploration of a lightweight, single-stage framework that uses compact depth features to reduce training computational cost and improve scalability for parallel training.

B. Motion Priors for Legged Locomotion

Motion priors have been widely adopted to regularize legged locomotion and alleviate the need for complex reward shaping. AMP-style methods have proven effective for inducing natural and agile behaviors [12], and have also enabled zero-shot generalization from flat-terrain motion datasets to more challenging terrains in quadruped locomotion [25]. Recent humanoid studies have incorporated motion priors to improve human-like locomotion characteristics [16] or provide fine-grained motion supervision through generative priors [17]. More recently, several works have investigated motion-prior-based humanoid locomotion on challenging terrains [22], [24], [26]. However, many existing methods continue to rely on flat-terrain motion priors that are not explicitly associated with specific terrain types, and adapt them to uneven terrains through staged training or additional residual mechanisms [24], [27]. Therefore, explicitly constructing terrain-related motion priors from demonstrations on target terrains remains underexplored in vision-based humanoid locomotion.

III. METHOD

The proposed framework is a lightweight single-stage, vision-assisted architecture for humanoid locomotion, which directly derives joint commands from raw depth images and onboard proprioception. By combining motion priors with a hierarchically structured perception module, the framework enables robust and human-like locomotion over complex terrains. An overview of the architecture and detailed explanations are presented below.

A. Overview

As shown in Fig. 2, our framework consists of three submodules: a hierarchical feature extractor, a terrain-related adversarial motion prior, and an asymmetric MoE actor-critic. The feature extractor learns latent representations of robot dynamics and environmental context from proprioceptive and exteroceptive observations, while explicitly estimating the base velocity $\hat{v}_t$. The motion prior guides the policy with flat-ground and stair locomotion demonstrations, encouraging terrain-adaptive and

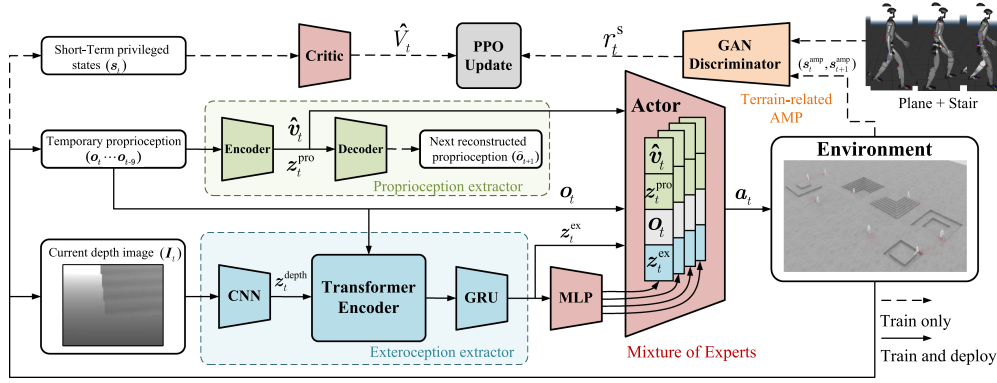


Fig. 2. Overview of the proposed framework. **Proprioception extractor (green)**: A β -variational autoencoder (β -VAE) processes a 10-frame window of proprioception $(o_{t-9:t})$ and implicitly captures the robot dynamics by performing one-step prediction of the next observation $\hat{o}_{t+1}$. It outputs a proprioceptive latent z_t^{pro} and an estimated body velocity $\hat{v}_t$. **Exteroception extractor (blue)**: A depth encoder compresses the current depth image I_t using a CNN, and produces an exteroceptive latent z_t^{ex} . **Motion-style guidance (yellow)**: An AMP discriminator evaluates the motion against a reference dataset and produces a motion style reward r_t^s , which is used exclusively during training. **Mixture-of-experts actor-critic (red)**: A MoE policy head receives $[\hat{v}_t, z_t^{\text{pro}}, o_t, z_t^{\text{ex}}]$ and outputs joint actions a_t , while the critic estimates the state-value function $\hat{V}_t$ for PPO.

human-like behaviors within a unified policy. To handle diverse terrain conditions, we adopt an MoE mechanism that enables terrain-aware behavior modulation. Additionally, the asymmetric actor-critic is designed for zero-shot transfer: the actor consumes only deployable observations, while the critic may leverage privileged states during training.

1) *MoE Policy Network*: To enable terrain-aware behavior modulation within a unified policy, we attach a MoE policy head to the actor. A gating network $g(\cdot)$ computes a convex combination of expert outputs based on the exteroceptive latent representation z_t^{ex} , producing the final action:

$$w_t = \text{softmax}(g(z_t^{\text{ex}})),$$

$$a_t = \sum_{k=1}^4 w_{t,k} \pi_k(x_t), \quad (1)$$

where the input x_t for each expert represents the concatenation of the current proprioceptive observation o_t , the VAE-derived history latent z_t^{pro} , the depth-derived exteroceptive latent z_t^{ex} , and the explicit base velocity estimate $\hat{v}_t$.

The proprioceptive observation $o_t \in \mathbb{R}^{96}$ is directly measured from the joint encoders and the IMU, and is defined as:

$$o_t = [\omega_t \quad g_t \quad c_t \quad \theta_t \quad \dot{\theta}_t \quad a_{t-1}]^T \quad (2)$$

which encompasses the body angular velocity ω_t , the gravity direction in the body frame g_t , the velocity command c_t , the joint angles θ_t , the joint velocities $\dot{\theta}_t$, and the previous action a_{t-1} .

2) *Value Network*: To obtain a more accurate estimate of the state value $\hat{V}_t$ during training, the critic is additionally provided with the privileged information unavailable at deployment. Its input vector is

$$s_t = [o_t \quad v_t \quad m_t^{\text{body}} \quad m_t^{\text{foot}}]^T, \quad (3)$$

TABLE I
REWARD FUNCTION ELEMENTS

Reward	Equation	Weight
Lin. velocity tracking	$\exp\{-4(v_{xy}^{\text{cmd}} - v_{xy})^2\}$	4.0
Ang. velocity tracking	$\exp\{-4(\omega_z^{\text{cmd}} - \omega_z)^2\}$	3.0
Linear velocity (z)	v_z^2	-1.0
Angular velocity (xy)	ω_{xy}^2	-0.05
Orientation	$ g ^2$	-1.0
Joint accelerations	$\dot{\theta}^2$	-2.5×10^{-7}
Joint power	$ \tau \dot{\theta} $	-1×10^{-3}
Action rate	$(a_t - a_{t-1})^2$	-0.03
Collision	$n_{\text{collision}}$	-1.0
Feet stumble	$\mathbb{1}_{\ F_{xy}^{\text{foot}}\ > 2\ F_z^{\text{foot}}\ }$	-2.0
Standstill	$\mathbb{1}_{\ v_{xy}\ < 0.1 \cdot \ \theta - \theta^{\text{stand}}\ }$	-1.0
Joint regularization	$\mathbb{1}_{\ v_{xy}\ > 0.1 \cdot \ \theta^{\text{hip}}\ ^2}$	-1.0

The symbol $\mathbb{1}$ denotes the indicator function.

where v_t is the ground-truth base velocity, $m_t^{\text{body}} \in \mathbb{R}^{17 \times 11}$ denotes the height map scan of the surrounding terrain with a grid resolution of 0.1m, and $m_t^{\text{foot}} \in \mathbb{R}^{5 \times 3}$ represents the local height map scan under the feet with a grid resolution of 0.05 m.

3) *Reward Function*: Designing effective reward functions requires substantial domain expertise, yet extensive tuning often produces only marginal improvements [12]. The uncertainty and variability of the terrain further exacerbate this challenge. Compared with the reward design in [15] for whole-body locomotion, our framework dispenses with extensive auxiliary coordination terms for the arms, waist, and feet. Instead, natural and humanlike motion patterns emerge from the AMP-based style reward r_t^s . Consequently, the remaining rewards are reduced to a compact set of fundamental motion-tracking and regularization terms [28], summarized in Table I.

B. Terrain-Related Motion Prior

To encourage terrain-adaptive motion patterns within a unified policy, we introduce a terrain-related adversarial motion prior.

Unlike conventional uses of AMP, which are often introduced to regularize generic locomotion style [16], [17] or built primarily on flat-ground motion data [24], [25], our reference dataset is constructed from flat-ground and stair locomotion demonstrations, so that the prior captures motion structures directly relevant to complex-terrain traversal.

We employ an MLP-based discriminator D_ϕ to distinguish whether a motion transition $\tau_s = (s_t^{\text{amp}}, s_{t+1}^{\text{amp}})$ comes from the reference dataset $\mathcal{D}$ or from agent rollouts $\mathcal{A}$. The motion-style state s_t^{amp} concatenates the base linear and angular velocities, joint positions and velocities, and end-effector offsets relative to the base in the base frame. The reference motions are derived from AMASS [29] and retargeted to the robot kinematic model using GMR [30]. In our implementation, the reference set is constructed from terrain-related flat-ground and stair locomotion demonstrations, with a total duration of approximately 30 s.

To account for the surface-contact characteristics of bipedal feet, we exclude the ankle-roll joint from the expert state, as including it was found to induce undesirable foot-ground misalignment. The discriminator is trained with the objective:

$$\begin{aligned} \arg \min_{\phi} \mathbb{E}_{\tau_s \sim \mathcal{D}} \left[(D_\phi(\tau_s) - 1)^2 \right] &+ \mathbb{E}_{\tau_s \sim \mathcal{A}} \left[(D_\phi(\tau_s) + 1)^2 \right] \\ &+ \frac{\alpha^{\text{gp}}}{2} \mathbb{E}_{\tau_s \sim \mathcal{D}} \left[\|\nabla_{\phi} D_\phi(\tau_s)\|_2 \right], \end{aligned} \quad (4)$$

where the first two terms follow the least-squares GAN objective and the last term is a gradient penalty weighted by $\alpha^{\text{gp}} = 10$. The resulting style reward is normalized to $[0, 1]$ as:

$$r_t^s = \max \left[0, 1 - 0.25 \times (d_t^{\text{score}} - 1)^2 \right], \quad (5)$$

where $d_t^{\text{score}} = D_\phi(\tau_s)$.

C. Hierarchical Feature Extractor

To capture the robot's internal state and environmental context, we employ a two-layer feature extractor. It consists of a proprioceptive extractor (PE) and an exteroceptive extractor (EE), enabling the robot to remain perceptive and adaptable in complex environments.

The first layer, PE, is responsible for learning the robot's short-horizon dynamics and estimating key motion variables from a 10-frame window of proprioceptive observations. It employs a β -VAE to encode a latent representation z_t^{pro} from the historical observations $\mathbf{o}_{t-9:t}$, while explicitly estimating the base velocity $\hat{\mathbf{v}}_t$. Following prior work [3], we adopt a β -VAE rather than a pure auto-encoder to regularize the latent space through a KL divergence term.

Instead of reconstructing the full successor observation $\hat{\mathbf{o}}_{t+1}$, the decoder only reconstructs the dynamics-related subset $\mathbf{o}_{t+1}^{\text{dyn}} = [\boldsymbol{\omega}_{t+1}, \mathbf{g}_{t+1}, \boldsymbol{\theta}_{t+1}, \dot{\boldsymbol{\theta}}_{t+1}]^T$, thereby excluding the command and previous-action terms. This design avoids imposing an artificial prediction target on future actions and focuses the latent representation on short-horizon proprioceptive dynamics.

The training loss for the β -VAE is:

$$\begin{aligned} \mathcal{L}_{VAE} = D_{\text{KL}}(q(z_t^{\text{pro}} | \mathbf{o}_{t-9:t}) \parallel p(z_t^{\text{pro}})) &+ \|\hat{\mathbf{v}}_t - \mathbf{v}_t\|_2 \\ &+ \left\| \hat{\mathbf{o}}_{t+1}^{\text{dyn}} - \mathbf{o}_{t+1}^{\text{dyn}} \right\|_2, \end{aligned} \quad (6)$$

where $q(z_t^{\text{pro}} | \mathbf{o}_{t-9:t})$ denotes the posterior distribution of z_t^{pro} conditioned on $\mathbf{o}_{t-9:t}$, and $p(z_t^{\text{pro}})$ is the prior distribution modeled as a standard normal distribution.

While the robot can perform blind locomotion using only PE [3], it often exhibits conservative behaviors, such as lifting the foot only after contacting a step. To address this limitation, the second layer, EE, provides anticipatory terrain cues from exteroceptive observations, thereby improving adaptation to challenging terrain variations. Specifically, a lightweight CNN first encodes the current depth image $\mathbf{I}_t$ into a compact geometric feature z_t^{depth} . This feature is then fused with the current proprioception $\mathbf{o}_t$ through a transformer-based module for cross-modal integration. To compensate for the limited field of view of the onboard camera, a GRU further aggregates temporally displaced terrain information, producing the exteroceptive latent representation z_t^{ex} , which is used by the gating network $g(\cdot)$.

Overall, this design forms a lightweight single-stage perception pipeline for vision-assisted humanoid locomotion. Compared with methods involving auxiliary reconstruction supervision [10], [18], depth-based terrain mapping [23], or more structured perception pipelines [7], our framework uses depth observations to learn a compact terrain representation for downstream control.

D. Training Details

1) *Simulation Platform*: We train our policy in Isaac Sim with 4,096 parallel environments on a single NVIDIA RTX 4090 GPU. All components are optimized in a single stage using the PPO [31] algorithm with a shared optimizer. To accelerate training, we replace the default camera with a ray-casting camera implemented via NVIDIA Warp, enabling faster depth image generation.

2) *Terrain Curriculum*: During training, the robot is exposed to a variety of challenging terrains, including stairs, slopes, high platforms, and gaps. Directly learning locomotion on such complex terrains is difficult; therefore, we employ a game-inspired curriculum [28] that gradually increases terrain difficulty based on the agent's policy performance. Within the curriculum, the slope increases from 0 to 0.4 rd, the step height from 0 to 0.4 m, the stair height from 0 to 0.23 m, and the gap width from 0.1 to 0.8 m.

3) *Early Termination*: Episodes are terminated early if any of the following conditions are met: excessive contact force on the torso link, body tilt exceeding 45 deg, base height falling below 0.5 m relative to the terrain, or timeout at the maximum episode length.

4) *Domain Randomization*: To improve policy robustness and facilitate the sim-to-real transfer, we introduce extensive domain randomization, covering variations in robot dynamics, physical properties, system latency, and camera parameters. The sampling ranges are summarized in Table II.

TABLE II
DOMAIN RANDOMIZATION RANGES

Parameter	Range	Unit
Friction	[0.4, 1.0]	/
Payload	[-5, 5]	kg
System delay	[0, 25]	ms
External push (xy)	[-1.0, 1.0]	m/s
Initial base yaw (z)	$[-\pi, \pi]$	rad
Initial base position (xy)	[-50, 50]	mm
Initial joint position	[0.5, 1.5]	rad
Camera pitch	[-2, 2]	deg
Camera position	[-5, 5]	mm
Depth Gaussian noise	[0, 50]	mm
Depth Gaussian filter kernel	[1, 3, 5]	/
Depth Gaussian filter sigma	[1.0, 1.0]	/
Depth deviation noise	[0, 10]	mm

IV. EXPERIMENTS

This section presents a comprehensive evaluation of the proposed framework. We conduct ablation studies and controlled comparisons with existing vision-assisted methods under unified settings, and further validate the learned policy through real-world deployment on indoor and outdoor challenging terrains. The following subsections describe the experimental setup, quantitative results, and analysis.

A. Experimental Setup

The proposed method is deployed and evaluated on a Unitree G1 humanoid robot (1.4 m tall and 35 kg). The robot is equipped with an Intel RealSense D435i camera, while all onboard computation is performed on a Jetson Orin NX. During hardware deployment, robot control and image acquisition are handled through ROS. Raw depth images are captured at 480×360 and 10 Hz, processed using built-in filters from the Intel RealSense API to reduce the sim-to-real gap, downsampled to 36×48 , and then cropped in the upper half, yielding a final 36×24 depth observation used as the policy input. The policy outputs target joint positions at 50 Hz, which are converted into actuation torques by a PD controller running at 500 Hz.

We compare the following ablation variants and baselines:

- **Ours w/o EE**: This method serves as a blind locomotion baseline [3], relying solely on proprioceptive inputs.
- **Ours w/o PE**: This method excludes the feature extraction from the temporal proprioceptive observations.
- **Ours w/o MoE**: This method is trained without the MoE mechanism.
- **Ours w/o TR-AMP**: This method excludes stair-related demonstrations.
- **MoRE** [24]: A two-stage method with flat-ground AMP.
- **AME-2** [23]: A unified RL method with depth-based terrain mapping and attention-based map encoding.

For a fair comparison, all policies are trained under the same configuration and evaluated in 2000 parallel environments over the same terrain suite, including stairs, gaps, and platforms. During evaluation, the commanded forward velocity is uniformly sampled from 0.7 to 1.0 m/s for all policies. The stair terrain contains both ascent and descent, with a step height of 20 cm

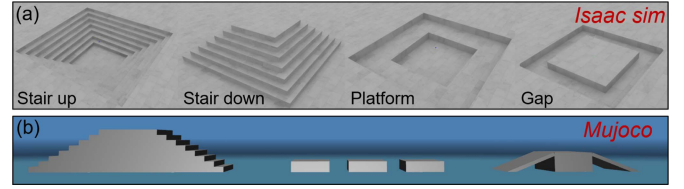


Fig. 3. Simulation terrains used in the experiments. (a) Representative Isaac Sim terrains. (b) Mixed terrain used for MuJoCo evaluation.

and a step depth of 30 cm. The platform terrain has a height of 40 cm, while the gap terrain has a width of 60 cm. We report four metrics: (i) success rate (SR), defined as the percentage of episodes in which the robot traverses the terrain without falling; (ii) average mechanical power consumption, computed as $P = \frac{1}{T} \sum_{t=1}^T \|\tau_t \odot \dot{q}_t\|_2$; (iii) the foot penetration penalty (FPP) at terrain edges, following [11]; and (iv) training time (TT), measured as the average wall-clock time per training iteration. In addition, we evaluate the traversability of all methods on continuous mixed terrain in MuJoCo over 20 trials. Fig. 3 illustrates the terrain used in the simulation experiment.

B. Simulation Experiment

1) *Performance Comparison*: Table III summarizes the performance of different methods across the four terrain configurations, together with their training efficiency.

Among the ablation variants, **Ours w/o EE** corresponds to blind locomotion without explicit terrain awareness, leading to low SR and high power consumption. Although **Ours w/o PE** maintains relatively high SR and low FPP, its power consumption increases slightly, suggesting less efficient motion generation. This highlights the role of PE in learning latent representations that capture robot dynamics and terrain-related factors. In addition, disabling either MoE or TR-AMP reduces SR and increases FPP, with a much more pronounced degradation observed for **Ours w/o TR-AMP**. The same variant also exhibits higher power consumption during stair ascent and descent, suggesting that TR-AMP helps the policy learn more efficient terrain-aware locomotion in such scenarios.

Compared with the external baselines, our method achieves the strongest overall trade-off among traversal performance, energy efficiency, and safety-related metrics. **MoRE** yields substantially lower SR than both AME-2 and our method, together with noticeably higher power consumption, suggesting that motion priors built mainly from flat-ground locomotion are less effective for complex-terrain behaviors. By contrast, both **AME-2** and our method achieve high SR across the four terrain configurations, indicating comparable traversal performance. However, our method consistently requires lower mechanical power. This suggests that, relative to the purely RL-based controller, the proposed TR-AMP-based framework produces more efficient locomotion while maintaining competitive traversal performance. Notably, both MoRE and AME-2 exhibit relatively high FPP, especially on stair descent and gap traversal, implying less stable foothold interactions in these demanding scenarios.

TABLE III
PERFORMANCE COMPARISON ACROSS FOUR TERRAIN CONFIGURATIONS WITH TRAINING EFFICIENCY

Method	Stairs Up			Stairs Down			Platform			Gap			Overall			MuJoCo	Sim
	SR (%) $\uparrow$	P (W) $\downarrow$	FPP $\downarrow$	SR	P	FPP	SR	P	FPP	SR	P	FPP	SR	P	FPP	SR	TT (s) $\downarrow$
w/o EE	36.9	131.6	0.95	47.7	114.2	2.23	0.55	155.4	—	0	—	—	21.3	133.7	1.59	0	1.88
w/o PE	98.1	114.3	0.30	98.7	89.5	0.54	96.5	90.2	0.24	96.4	87.5	0.08	97.4	95.4	0.29	95	2.73
w/o MoE	96.4	110.1	0.41	98.1	86.4	0.74	88.1	86.9	0.28	93.8	85.1	0.16	94.1	92.1	0.40	90	2.79
w/o TR-AMP	92.3	120.7	0.56	88.6	97.3	1.12	85.5	88.7	0.29	92.4	87.9	0.28	89.7	98.7	0.56	80	2.85
MoRE [24]	91.6	159.6	0.74	92.8	161.5	2.44	43.3	150.1	0.43	95.9	125.2	0.21	80.9	149.1	0.96	65	11.35
AME-2 [23]	98.7	131.6	0.62	99.2	129.1	2.80	98.6	107.2	0.36	96.3	100.1	0.56	98.2	117.0	1.09	95	7.68
Ours	98.4	108.4	0.25	99.6	85.2	0.53	98.2	86.2	0.23	97.5	84.6	0.04	98.4	91.1	0.26	100	2.85

— indicates unavailable or not applicable results. The best results are in **bold**, and $\downarrow$ indicates that lower values are better. The training efficiency result of TT is reported on an NVIDIA RTX 4090 GPU.

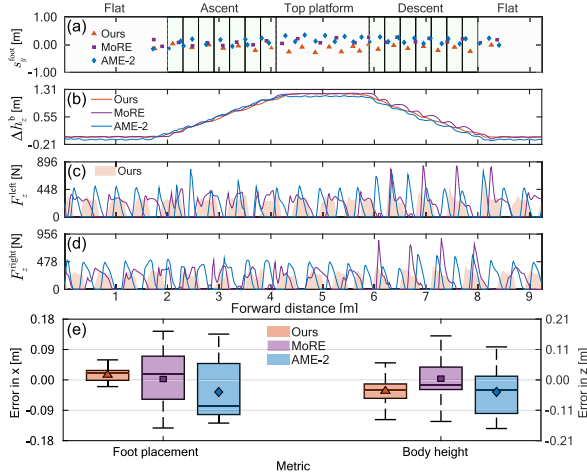


Fig. 4. Comparison of stair-traversal behaviors among different methods. (a) Top-view foot placement. (b) Body-height variation relative to the forward distance. (c)–(d) Vertical ground reaction forces of both feet. (e) Box plots of foot-placement and body-height errors.

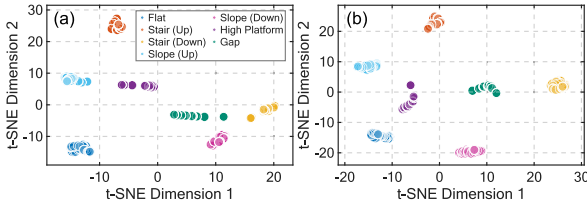


Fig. 5. Visualization of the depth-derived exteroceptive latent space z_t^{ex} for (a) our method without MoE and (b) the proposed method, using 1 s trajectories collected from each of the seven terrain types.

In terms of training efficiency, our method also shows clear advantages, with a substantially lower average training time per iteration than the baselines. This efficiency gain is consistent with our lightweight perception pipeline, which directly extracts compact terrain cues from depth observations without relying on more structured map encoding or multi-stage training procedures.

2) *Stair Traversal Behavior Analysis:* To further evaluate the effect of the proposed TR-AMP on terrain-dependent locomotion behavior, we compare our method with MoRE and AME-2 on a continuous stair traversal task in MuJoCo, where the robot performs stair ascent, level walking, and stair descent at a desired

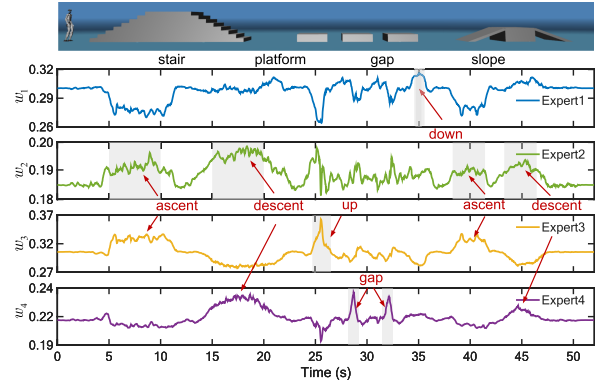


Fig. 6. MoE expert activation over a mixed-terrain trajectory, represented by the gating weights of the four experts.

forward speed of 0.8 m/s. Fig. 4 provides both trajectory-level and statistical comparisons of stair traversal behavior.

Compared with the baselines, our method exhibits more consistent stair-traversal behavior. As shown in Fig. 4(a)–(b), the foot placements are distributed closer to the stair midlines and the body-height variation is smaller, especially during stair descent, indicating more accurate foothold selection and improved whole-body stability across terrain transitions. The box plots in Fig. 4(e) further confirm smaller foot-placement and body-height errors. Meanwhile, Fig. 4(c)–(d) shows generally lower and less impulsive vertical ground reaction forces for both feet, suggesting more compliant foot–ground interactions during stair traversal.

These results indicate that the proposed TR-AMP enables the policy to invoke terrain-compatible locomotion modes under visual guidance, rather than relying on a single generic gait pattern across all terrain phases. This benefit is particularly evident during stair descent, where the learned adaptation yields more stable body motion, more suitable foothold placement, and lower contact forces.

3) *Latent Space and Expert Activation Analysis:* Fig. 5 shows the t-SNE visualization of the depth-derived exteroceptive embeddings. Without MoE, the terrain clusters are less clearly separated. After introducing MoE, the embeddings exhibit more distinct cluster boundaries and stronger inter-terrain separation, suggesting that the learned visual features become more discriminative across terrain types.

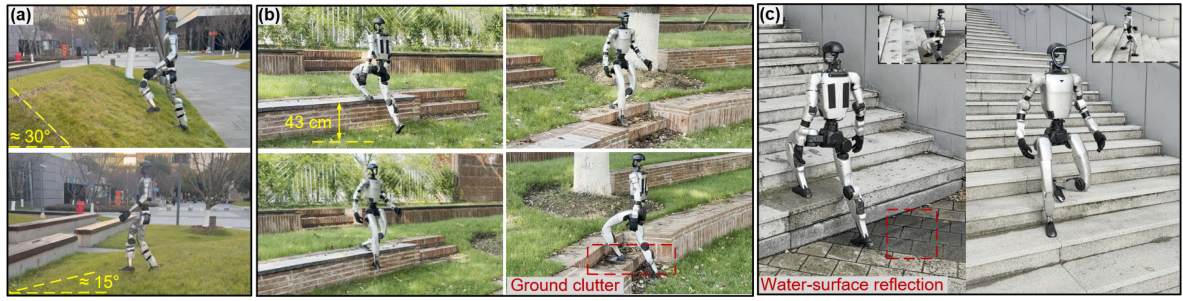


Fig. 7. Representative real-world rollouts of our method in diverse deployment scenes. (a) Outdoor grassy slopes. (b) Mixed outdoor terrain with stair-like transitions and elevated platforms under natural ground clutter. (c) Repeated ascent and descent on a long staircase with 46 consecutive steps, with water-surface reflection causing depth degradation.

To further clarify how these terrain clusters relate to the four experts, we track the gating weights along a mixed-terrain trajectory containing stair ascent/descent, platform stepping, gap crossing, and slope ascent/descent, as shown in Fig. 6. The results reveal clear but non-exclusive expert specialization. In particular, Expert 2 consistently receives high weights on stairs and slopes, suggesting that it captures control modes associated with continuous elevation changes. Expert 3 is most strongly activated when stepping onto the platform, while Expert 4 is primarily activated during gap traversal. By contrast, Expert 1 dominates on relatively flat segments and is suppressed when the robot encounters more challenging terrain regions. Importantly, the same expert can be reused across multiple related terrains. For example, experts activated on stairs may also respond to slopes. This indicates that the four experts are not assigned to terrain categories in a one-to-one manner. Instead, the gating network organizes them into reusable control modes, allowing multiple fine-grained terrain clusters to be handled by a smaller set of specialized experts.

C. Real-World Experiments

We validate the zero-shot sim-to-real transfer of the proposed method in a variety of indoor and outdoor environments. In indoor experiments, the robot successfully performs repeated traversals over three representative terrain types, including stairs, platforms, and gaps, as shown in Fig. 1. Specifically, we conduct 20 consecutive trials on indoor stairs with a step height of 15 cm and a step depth of 30 cm, where each trial consists of a complete ascent–descent traversal. In addition, the robot completes 20 consecutive trials on a 30 cm platform and a 70 cm gap. The success rate is 100% on all three standardized indoor tasks, demonstrating reliable deployment without task-specific fine-tuning.

We further evaluate the method in more challenging outdoor scenes, including a 30° slope, a 43 cm platform, and a long staircase with 46 consecutive steps, as illustrated in Fig. 7. On the staircase, the robot successfully performs both ascent and descent. These outdoor environments also contain substantial visual disturbances, such as grass, fallen leaves, twigs, and a puddle before the stairs, which causes reflective degradation in the depth observations. Despite these factors, the policy remains robust and maintains stable locomotion behavior across all tested

TABLE IV
COMPARISON OF COMPREHENSIVE CAPABILITIES

Method	Robot	Representation	Long-stair	Platform	Gap
HPL [11]	Unitree-H1	Vision	No	0.48×	0.93×
PIM [15]	Unitree-H1	Elevation Map	No	0.57×	0.57×
Gallant [32]	Unitree-G1	Voxel	No	0.46×	0.61×
STS [21]	Limx Oli	Vision-to-Elevation	Yes	—	0.5×
Ours	Unitree-G1	Vision	Yes	0.66×	1.07×

The symbol × indicates the value normalized by the robot leg length.

terrains. Notably, the learned policy exhibits terrain-aware motion characteristics consistent with the introduced TR-AMP. For example, during stair descent, the robot actively lowers its center of mass to improve stability, rather than stepping down with an unchanged posture.

D. Discussion

As summarized in Table IV, compared with existing perceptive humanoid locomotion systems, our approach achieves a more balanced performance across different challenging terrains, maintaining stable long-stair traversal while remaining competitive on tasks such as high-platform stepping and wide-gap crossing.

The merit of the proposed method is that it introduces terrain-relevant motion regularity into direct vision-based humanoid locomotion, enabling the policy to adapt its locomotion style across terrains rather than relying on a single generic gait pattern. This benefit is reflected not only in traversal performance, but also in the quality of the learned motion, including improved energy efficiency, more safety-related contact behavior, and more stable stair negotiation. More broadly, the learned policy transfers robustly to cluttered real-world scenes with reflections, vegetation, and long continuous stair sequences, while preserving terrain-adaptive locomotion behavior. These findings suggest that terrain-related motion priors provide an effective inductive bias for coupling visual perception with locomotion control in complex-terrain humanoid deployment.

However, our approach may be less effective on highly discrete terrains, such as stilt-like obstacles [23], where precise foothold selection requires finer-grained geometric information. In these cases, the attention-based method in [23] can offer stronger performance, but with lower training efficiency and

higher computational cost. This trade-off reflects the different design emphasis between structured map-based perception and lightweight latent terrain encoding.

V. CONCLUSION

In this work, we present a lightweight single-stage vision-assisted learning framework for bipedal locomotion that integrates a hierarchical feature extractor with an MoE actor, enabling terrain-aware control across diverse terrain conditions. By further introducing a terrain-related adversarial motion prior, the proposed method encourages terrain-adaptive and human-like locomotion behaviors without relying on multi-stage training. Extensive simulation and real-world experiments demonstrate robust generalization across a wide range of indoor and outdoor environments, including stairs, platforms, gaps, and slopes.

In future work, we plan to evaluate the proposed approach on more challenging terrains, such as discrete stilts. In addition, obstacle avoidance during agile locomotion will be an important direction for further investigation.

AAPPENDIX A DETAILED NETWORK ARCHITECTURE

Table V provides the implementation details of the perception

TABLE V
DETAILED I/O AND PARAMETERS OF ALL COMPONENTS

EE	Depth Encoder	$z_t^{\text{depth}} = \text{MLP}(\text{CNN}(I_t))$ $\text{CNN: } \{[32,64,64],[4,3,2],[3,3,2],[3,2,1]\}$ $\text{MLP: } \{[128], \text{ELU}\}$
	Fusion	$z_t^{\text{ff}} = L(\text{SA}(\mathbf{o}_t, z_t^{\text{depth}}))$ $\text{SA: } \{[256,256],64,1\}, L: \text{Linear}$
	Temporal Encoder	$z_t^{\text{ex}} = \text{GRU}(z_t^{\text{ff}})$ $\text{GRU: } \{1,128\}$
VAE	Encoder	$z_t^{\text{pro}}, \hat{\mathbf{v}}_t = \text{MLP}(\mathbf{o}_t, \mathbf{v}_{t-1:t})$ $\text{MLP: } \{[128,64], \text{ELU}\}$
	Decoder	$\hat{\mathbf{o}}_{t+1} = \text{MLP}(z_t^{\text{pro}})$ $\text{MLP: } \{[64,128], \text{ELU}\}$
MoE	Experts	$a_t^{(i)} = \text{MLP}([\mathbf{o}_t, \hat{\mathbf{v}}_t, z_t^{\text{pro}}, z_t^{\text{depth}}]), i = 1, \dots, 4$ 4 experts, $\text{MLP: } \{[512,256,128], \text{ELU}\}$
	Gate	$\mathbf{w}_t = \text{Softmax}(\text{MLP}(z_t^{\text{ex}}))$ $\text{MLP: } \{[64], \text{ELU}\}$
Critic	Value	$V_t = \text{MLP}(s_t)$ $\text{MLP: } \{[512,256,128], \text{ELU}\}$
$\text{CNN: } \{\text{channel, kernel, stride, padding}\}, \text{MLP: } \{\text{hidden, activation}\}$ $\text{GRU: } \{\text{n_layer, hidden}\}, \text{SA: } \{\text{hidden, token, n_head}\}$		

encoder, proprioceptive encoder, mixture-of-experts policy, and critic.

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